

Homework (Jalapeno)

- Q1.** In ancient Egypt, a laborer built $\frac{1}{4}$ of a pyramid in 5 days. What is the laborer's rate of building pyramids per day?
- (A) $\frac{1}{20}$
(B) $\frac{1}{4}$
(C) $\frac{1}{5}$
(D) $\frac{1}{10}$
(E) $\frac{1}{2}$
- Q2.** The Great Wall of China has an average height of 7 meters. If a section of the wall is 35 meters long, what is the ratio of the wall's height to its length?
- (A) 1 : 5
(B) 1 : 10
(C) 1 : 7
(D) 1 : 20
(E) 1 : 25
- Q3.** A group of archaeologists discovered an artifact that is 3 times as old as the pyramids of Giza. If the pyramids are approximately 4500 years old, what is the age of the artifact?
- (A) 13500 years
(B) 12000 years
(C) 10500 years
(D) 9000 years
(E) 7500 years
- Q4.** The rate at which the Roman Empire expanded its territories is directly proportional to the number of legions it had. If 5 legions expanded the empire by 2000 square miles, what is the rate of expansion per legion?
- (A) 400 square miles per legion
(B) 500 square miles per legion
(C) 600 square miles per legion
(D) 800 square miles per legion
(E) 1000 square miles per legion
- Q5.** The cost of excavating an archaeological site is inversely proportional to the number of workers. If 10 workers can excavate a site for 1200, what is the cost per worker?
- (A) 120
(B) 150
(C) 180
(D) 200
(E) 250
- Q6.** The ancient Greeks built a water clock that flowed at a rate of 2 liters per hour. How many liters would the clock flow in 5 hours?
- (A) 8 liters
(B) 10 liters
(C) 12 liters
(D) 15 liters
(E) 20 liters
- Q7.** A museum has a collection of ancient vases that are valued at 5000 each. If the collection contains 15 vases, what is the total value of the collection?
- (A) 50000
(B) 60000
(C) 70000
(D) 75000
(E) 80000
- Q8.** The speed of a chariot in ancient Rome is directly proportional to the number of horses pulling it. If a chariot with 4 horses can travel 20 miles per hour, what is the speed of a chariot with 6 horses?
- (A) 25 miles per hour
(B) 30 miles per hour
(C) 35 miles per hour
(D) 40 miles per hour
(E) 45 miles per hour

Open Ended Questions (FRQ)

- Q9.** The ancient city of Pompeii was destroyed by the eruption of Mount Vesuvius in 79 AD. The city's population was approximately 20000 people. If the population grew at a rate of 2% per year, what would have been the population of Pompeii 10 years after the eruption if it had not been destroyed?
- Q10.** A historian is studying the construction of the Great Pyramid of Giza. The pyramid's original height was approximately 146 meters. If the pyramid's height is reduced by 0.5% every year due to erosion, what will be the height of the pyramid 50 years from now?

Chef's Tips

- Tip 1: Ensure students show all work and calculations for free response questions
- Tip 2: Encourage students to use units and labels in their answers
- Tip 3: Remind students to check their work for accuracy and completeness